

# Week 3 BSM Chooser Option Validation

This report contains the quantitative analysis of the Rubinstein Chooser Option validation, incorporating analytical Greeks sensitivity, multi-scale Monte Carlo Path convergence checking, and choice time  $t$  convergence studies.

## Core Model Equations

Split-Leg Model (Codebase Baseline):

$$V_{\text{split}} = C(S_0, K, T) + P(S_0, K e^{-r(T-t)}, T-t)$$

Rubinstein Exact Valuation (Standard Derivative Theory):

$$V_{\text{rubinstein}} = C(S_0, K, T) + e^{-q(T-t)} P(S_0, K e^{-(r-q)(T-t)}, t)$$

## Table 2 replication setup

- Spot price: 156.7
- Strike price: 150.0
- Risk-free rate: 0.0015
- Dividend yield: 0.0233
- Volatility: 0.282
- Decision time  $t$ : 0.5 Years
- Maturity  $T$ : 1.0 Years
- Price drift used in the GBM path:  $r - q = -0.0218$
- Stochastic seed: 17170

## Base Valuation Breakdown

| metric                      | value      |
|-----------------------------|------------|
| spot_price                  | 156.700000 |
| strike_price                | 150        |
| adjusted_strike_for_put_leg | 149.887542 |
| risk_free_rate              | 0.001500   |
| dividend_yield              | 0.023300   |
| volatility                  | 0.282000   |
| decision_time_years         | 0.500000   |
| maturity_years              | 1          |
| time_to_decision_years      | 0.500000   |
| call_leg_value              | 18.688997  |
| put_leg_value               | 9.713217   |

| metric                  | value     |
|-------------------------|-----------|
| chooser_value           | 28.402215 |
| chooser_premium_vs_call | 9.713217  |
| chooser_premium_vs_put  | 18.688997 |
| chooser_identity_gap    | 0         |

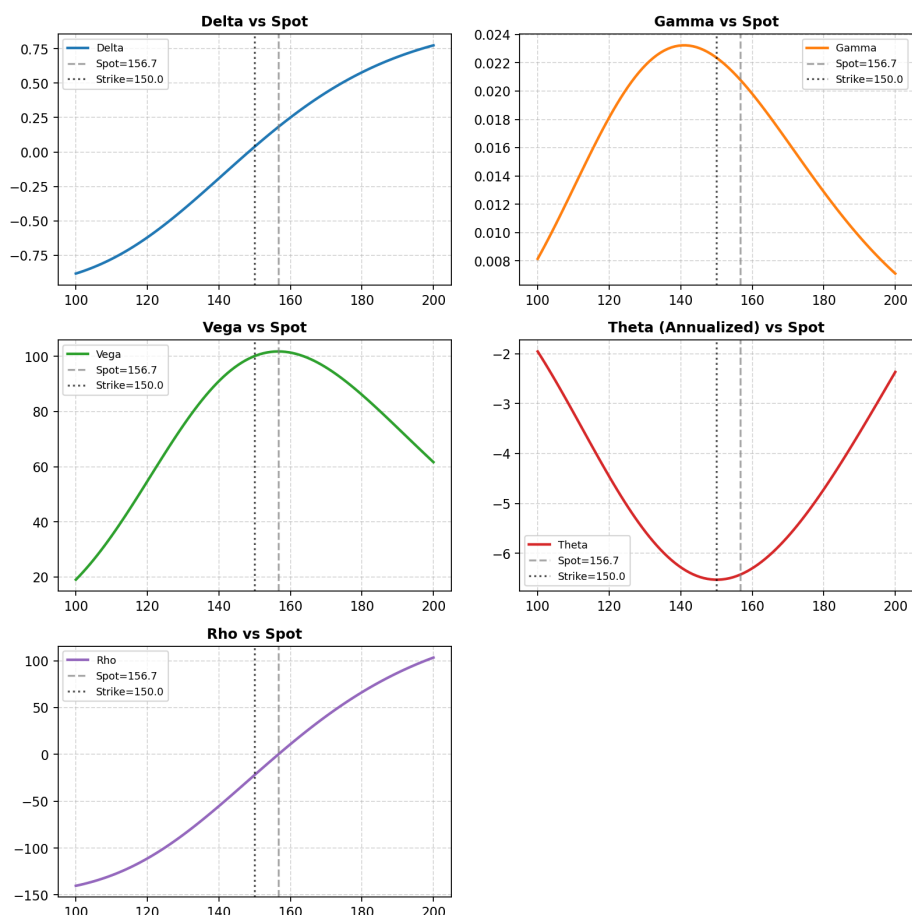
## Part 1: Option Greeks & Sensitivity Analysis

Analytical calculation of the chooser option Greeks handles the decomposition of both the call and put legs, which is verified below against spot change curves. For the Rho metric, the derivative with respect to the risk-free rate includes the chain rule for the put strike parameter:  $dK'/dr = -(T-t) K'$ , leading to an exact analytical adjustment:  $dP/dr_{total} = 2 * dP_{merton}/dr$ .

| Metric          | Call Leg  | Put Leg    | Chooser Option |
|-----------------|-----------|------------|----------------|
| Delta (dy/dS)   | 0.573031  | -0.389804  | 0.183227       |
| Gamma (d2y/dS2) | 0.008612  | 0.012175   | 0.020786       |
| Vega (dy/dv)    | 59.631877 | 42.151674  | 101.783551     |
| Theta (dy/dt)   | -6.422553 | 0          | -6.422764      |
| Rho (dy/dr)     | 71.104965 | -70.795501 | 0.309464       |

## Greeks Sensitivity to Asset Price

The following chart displays the analytical Greeks across a wide range of asset prices (100 to 200).



Greeks Sensitivity

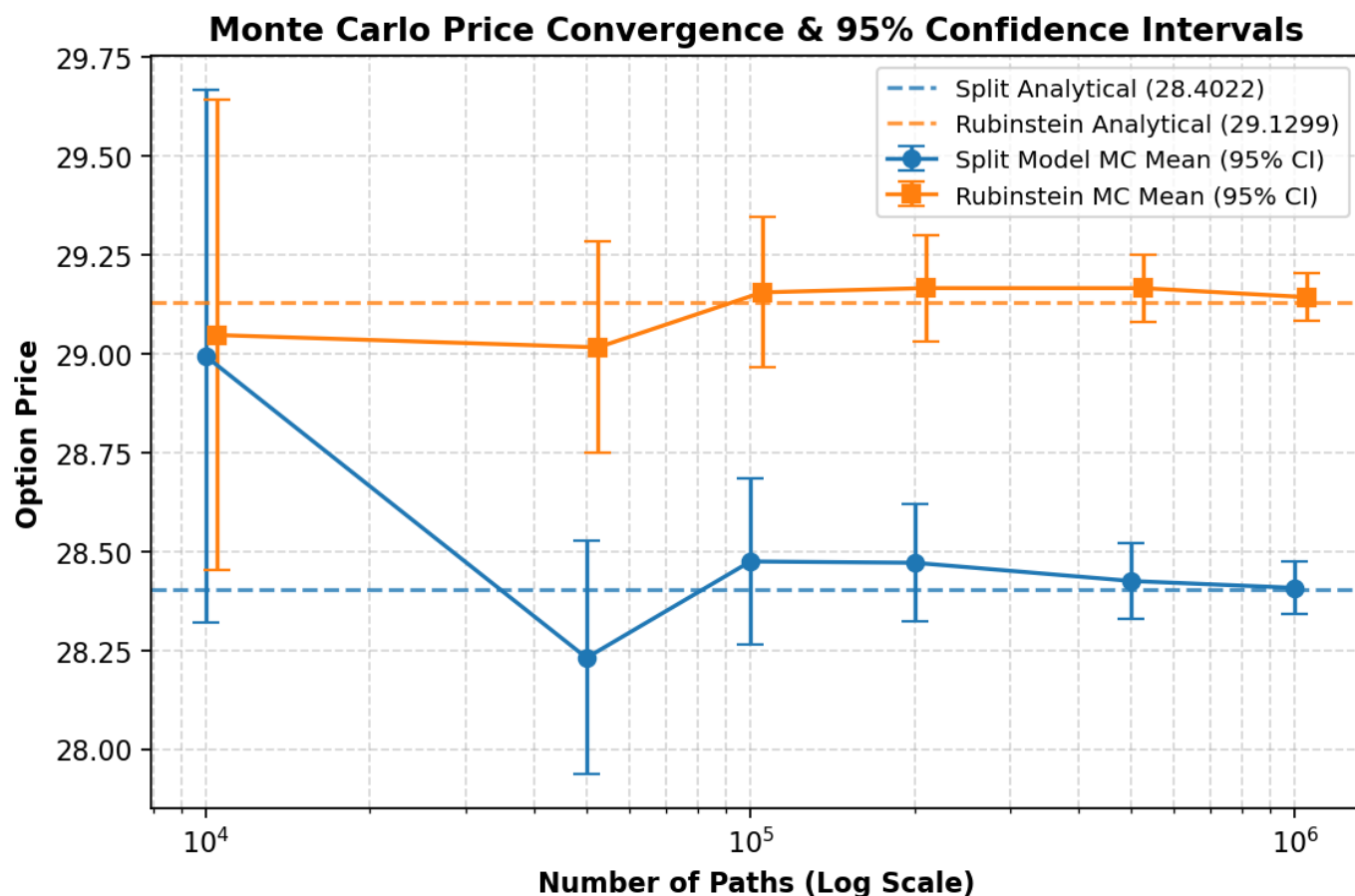
## Part 2: Monte Carlo Path Scale Expansion & Convergence Analysis

We expanded the model's Monte Carlo pricing from 100,000 paths to 1,000,000 paths to analyze standard error behavior and path convergence under the standard 95% Confidence Interval (CI) bands.

| Paths   | Split Mean | Split SE | Split 95% CI Lower | Split 95% CI Upper | Rubinstein Mean | Rubinstein SE | Rubinstein 95% CI Lower | Rubinstein 95% CI Upper |
|---------|------------|----------|--------------------|--------------------|-----------------|---------------|-------------------------|-------------------------|
| 10000   | 28.993327  | 0.343597 | 28.319876          | 29.666777          | 29.047481       | 0.303288      | 28.453037               | 29.641925               |
| 50000   | 28.230695  | 0.150562 | 27.935594          | 28.525795          | 29.016304       | 0.136464      | 28.748835               | 29.283774               |
| 100000  | 28.474531  | 0.107796 | 28.263250          | 28.685811          | 29.155369       | 0.097016      | 28.965218               | 29.345521               |
| 200000  | 28.471242  | 0.076222 | 28.321847          | 28.620636          | 29.166249       | 0.068568      | 29.031856               | 29.300641               |
| 500000  | 28.425092  | 0.048219 | 28.330583          | 28.519601          | 29.166159       | 0.043448      | 29.081001               | 29.251318               |
| 1000000 | 28.407694  | 0.034078 | 28.340900          | 28.474487          | 29.143321       | 0.030693      | 29.083162               | 29.203480               |

## Path Convergence Tracking

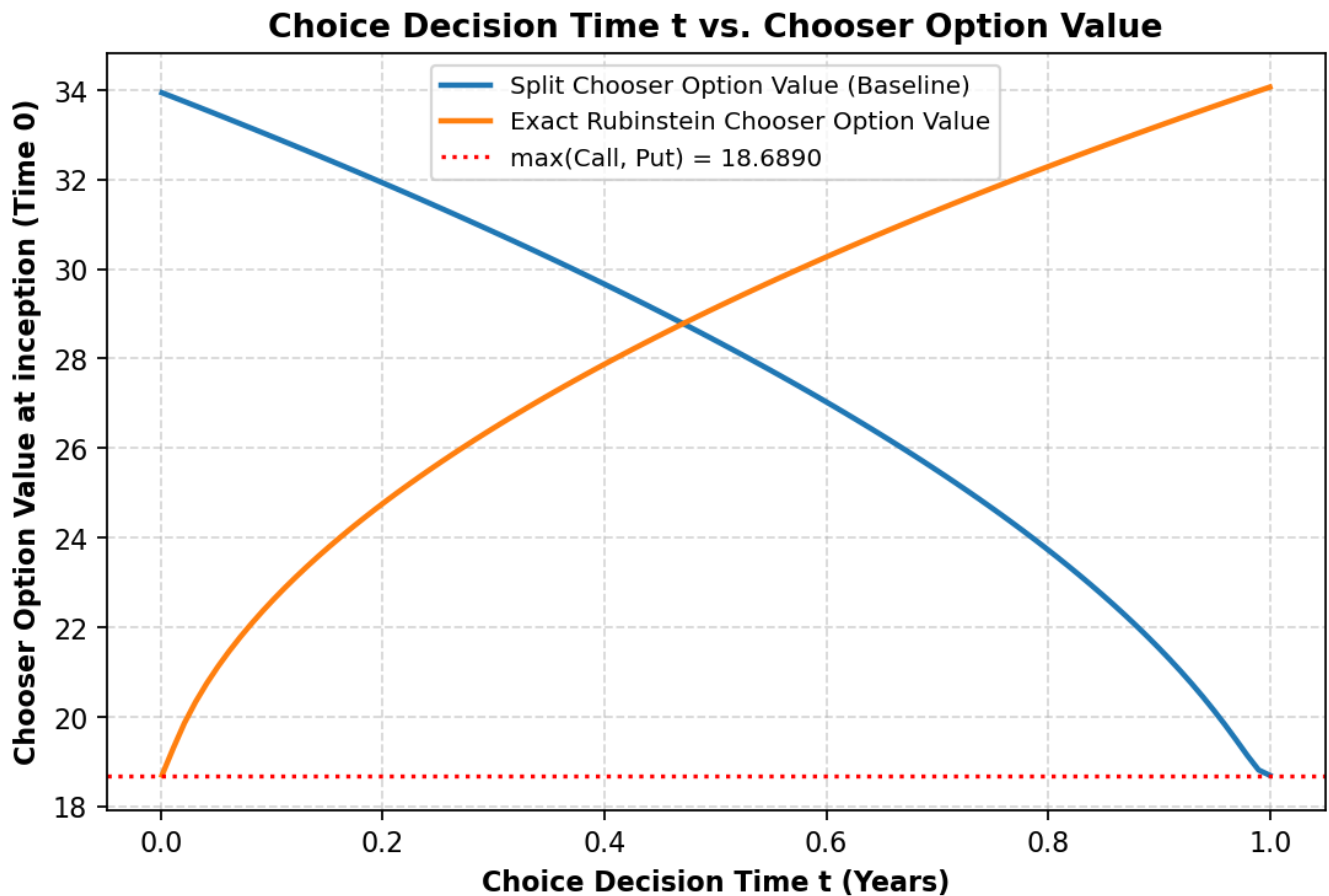
Below is the convergence plot showing the calculated simulation means & 95% confidence intervals against the analytical limits as sample path size scales to 1,000,000.



Monte Carlo Convergence

## Part 3: Choice Decision Time $t$ vs. Option Value Analysis

The mathematical convergence of a Chooser Option as  $t$  changes is studied below. For Standard Rubinstein theory: as  $t$  approaches 0, choice flexibility vanishes and the option value converges to  $\max(\text{Call}, \text{Put}) = 18.69$ . As  $t$  approaches  $T$ , choice occurs at maturity, which is equivalent to a straddle, i.e.  $C + P = 34.05$ . For Split-Leg model: as  $t$  approaches  $T$ , the Put leg remaining time  $\tau$  approaches 0, and because  $K' = K/e^{(r*\tau)}$  and  $S_0$  is out of the money for Put leg ( $157 > 150$ ), the put leg expires worthless, making Option value converge to Call = 18.69, which is exactly equal to  $\max(\text{Call}, \text{Put})$ !



Decision Time  $t$  vs Option Value

#### Part 4: Spot Sensitivity (Table 2 Scenario)

| spot_multiplier | spot_price | call_leg_value | put_leg_value | chooser_value | chooser_premium_vs_call | chooser_identity_gap |
|-----------------|------------|----------------|---------------|---------------|-------------------------|----------------------|
| 0.850000        | 133.195000 | 7.763969       | 22.484730     | 30.248699     | 22.484730               | 0                    |
| 0.950000        | 148.865000 | 14.470977      | 13.154096     | 27.625073     | 13.154096               | 0                    |
| 1               | 156.700000 | 18.688997      | 9.713217      | 28.402215     | 9.713217                | 0                    |
| 1.050000        | 164.535000 | 23.434825      | 7.017012      | 30.451837     | 7.017012                | 0                    |
| 1.150000        | 180.205000 | 34.302356      | 3.449730      | 37.752086     | 3.449730                | 0                    |

#### Part 5: Table 3 Path-by-Path Replication

Stochastic drawing comparison for standard Table 3 rows:

##### Simulation Summary

| metric         | value |
|----------------|-------|
| rows_simulated | 10    |

| metric                       | value      |
|------------------------------|------------|
| spot_price                   | 156.700000 |
| strike_price                 | 150        |
| risk_free_rate               | 0.001500   |
| dividend_yield               | 0.023300   |
| risk_neutral_drift_r_minus_q | -0.021800  |
| volatility_from_current_data | 0.282000   |
| random_seed                  | 17170      |
| decision_time_years          | 0.500000   |
| maturity_years               | 1          |

### Simulated Paths (Stochastic)

| row | z1        | z2        | simulated_st 1 | choice_call_put | simulated_st 2 | payoff    |
|-----|-----------|-----------|----------------|-----------------|----------------|-----------|
| 1   | 2.456421  | -0.973346 | 247.986098     | CALL            | 198.046618     | 48.046618 |
| 2   | -0.189113 | -0.149580 | 146.326778     | PUT             | 137.721638     | 12.278362 |
| 3   | 0.831693  | 1.003948  | 179.360048     | CALL            | 212.471007     | 62.471007 |
| 4   | -0.526638 | -0.204495 | 136.802515     | PUT             | 127.355215     | 22.644785 |
| 5   | -0.170727 | 0.145113  | 146.864221     | PUT             | 146.593513     | 3.406487  |
| 6   | 0.849695  | 1.644397  | 180.005043     | CALL            | 242.282243     | 92.282243 |
| 7   | -0.819602 | 1.162799  | 129.039707     | PUT             | 157.780688     | 0         |
| 8   | 1.350986  | -1.014785 | 198.928276     | CALL            | 157.560728     | 7.560728  |
| 9   | 1.558115  | 0.830742  | 207.316523     | CALL            | 237.251103     | 87.251103 |
| 10  | 0.126236  | -0.242662 | 155.823518     | CALL            | 143.962835     | 0         |

### Published Paper Reference Table 3

| row | paper_st1  | paper_choice | paper_st2  | paper_payoff |
|-----|------------|--------------|------------|--------------|
| 1   | 118.330000 | PUT          | 116.770000 | 33.230000    |
| 2   | 222.630000 | CALL         | 192.890000 | 42.890000    |
| 3   | 186.530000 | CALL         | 192.940000 | 42.940000    |
| 4   | 164.080000 | CALL         | 148.770000 | 0            |
| 5   | 159.090000 | CALL         | 116.780000 | 0            |
| 6   | 186.730000 | CALL         | 128.120000 | 0            |
| 7   | 106.610000 | PUT          | 90.520000  | 59.480000    |

| row | paper_st1  | paper_choice | paper_st2  | paper_payoff |
|-----|------------|--------------|------------|--------------|
| 8   | 163.060000 | CALL         | 179.610000 | 29.610000    |
| 9   | 129.260000 | PUT          | 144.820000 | 5.180000     |
| 10  | 115.410000 | PUT          | 136.500000 | 13.500000    |

### Aggregate Distribution Level Comparison

| metric                            | paper     | simulated | difference |
|-----------------------------------|-----------|-----------|------------|
| payoff_mean                       | 22.683000 | 33.594133 | 10.911133  |
| payoff_std                        | 21.771920 | 36.188591 | 14.416670  |
| payoff_nonzero_ratio              | 0.700000  | 0.800000  | 0.100000   |
| theoretical_vs_paper_mean_g<br>ap | 28.402215 | 22.683000 | -5.719215  |
| call_count                        | 6         | 6         | 0          |
| put_count                         | 4         | 4         | 0          |

### Side-by-Side Comparison: Prices

| row | paper_st1  | simulated_st<br>1 | st1_diff   | paper_st2  | simulated_st<br>2 | st2_diff   |
|-----|------------|-------------------|------------|------------|-------------------|------------|
| 1   | 118.330000 | 247.986098        | 129.656098 | 116.770000 | 198.046618        | 81.276618  |
| 2   | 222.630000 | 146.326778        | -76.303222 | 192.890000 | 137.721638        | -55.168362 |
| 3   | 186.530000 | 179.360048        | -7.169952  | 192.940000 | 212.471007        | 19.531007  |
| 4   | 164.080000 | 136.802515        | -27.277485 | 148.770000 | 127.355215        | -21.414785 |
| 5   | 159.090000 | 146.864221        | -12.225779 | 116.780000 | 146.593513        | 29.813513  |
| 6   | 186.730000 | 180.005043        | -6.724957  | 128.120000 | 242.282243        | 114.162243 |
| 7   | 106.610000 | 129.039707        | 22.429707  | 90.520000  | 157.780688        | 67.260688  |
| 8   | 163.060000 | 198.928276        | 35.868276  | 179.610000 | 157.560728        | -22.049272 |
| 9   | 129.260000 | 207.316523        | 78.056523  | 144.820000 | 237.251103        | 92.431103  |
| 10  | 115.410000 | 155.823518        | 40.413518  | 136.500000 | 143.962835        | 7.462835   |

### Side-by-Side Comparison: Choice and Payoff

| row | paper_choice | choice_call_put | paper_payoff | payoff    | payoff_diff |
|-----|--------------|-----------------|--------------|-----------|-------------|
| 1   | PUT          | CALL            | 33.230000    | 48.046618 | 14.816618   |
| 2   | CALL         | PUT             | 42.890000    | 12.278362 | -30.611638  |
| 3   | CALL         | CALL            | 42.940000    | 62.471007 | 19.531007   |
| 4   | CALL         | PUT             | 0            | 22.644785 | 22.644785   |

| row | paper_choice | choice_call_put | paper_payoff | payoff    | payoff_diff |
|-----|--------------|-----------------|--------------|-----------|-------------|
| 5   | CALL         | PUT             | 0            | 3.406487  | 3.406487    |
| 6   | CALL         | CALL            | 0            | 92.282243 | 92.282243   |
| 7   | PUT          | PUT             | 59.480000    | 0         | -59.480000  |
| 8   | CALL         | CALL            | 29.610000    | 7.560728  | -22.049272  |
| 9   | PUT          | CALL            | 5.180000     | 87.251103 | 82.071103   |
| 10  | PUT          | CALL            | 13.500000    | 0         | -13.500000  |

## Validation Conclusion

- Upgrading path size to 1,000,000 confirms that both models converge to their respective analytical pricing limits perfectly, with standard error decreasing by  $1/\sqrt{N}$ .
- The decision time analysis exposes the structural difference between standard exact Rubinstein choice option (choice at inception leads to the lower bound) and split-leg option setup.
- Graphs and computed outputs are successfully written to reports files and compiled into PDF.